

Project information

Name	The ALMA survey to Resolve exoKuiper belt Substructures (ARKS)
Code	2022.1.00338.L
PI	Sebastian Marino
Organization	Institute of Astronomy, Cambridge, University of
Co-Is	P. Abraham, M. Booth, A. Brennan, J. Carpenter, G. Cataldi, E. Chiang, E. Choquet, C. Davies, R. Dawson, C. del Burgo, B. Dent, S. Ertel, T. Henning, A. Higuchi, A. Hughes, M. Jankovic, J. Jennings, A. Kospal, A. Krivov, S. Lawler, J. Lestrade, T. Loehne, J. Lovell, M. MacGregor, J. Marshall, L. Matra, B. Matthews, J. Milli, T. MIZUKI, M. Momose, A. Moor, R. Nakatani, K. Oberg, S. Ohashi, R. Orihara, N. Pawellek, J. Rigley, A. Sefilian, A. Sepulveda, D. Wilner, M. Wyatt

ObsUnitSet information

Name	Member OUS (HD14055)
QA2 Status	🟢SemiPass
Member OUS Status ID	uid://A001/X2d20/X2e71
SchedBlock name	HD14055_a_07_7M
SchedBlock UID	uid://A001/X2d20/X2e17
Array	7M
Mode	Pipelineable
Band	ALMA_RB_07
Repr.Freq. (sky)	345.14 [GHz]
Spectral setup	ACA
Sources	HD14055
Other SBs in this Group	HD14055_a_07_TM2 (uid://A001/X2d20/X2e6f), HD14055_a_07_TM1
OUS (Member OUS)	(uid://A001/X2d20/X2e6d)
Status ID in brackets):	
Execution count	21.56 of 22 expected

Final QA2 comment

Data reduction method(s): PipelineCalAndImaging

Calibration issues: In EB uid://A002/X1006d51/X228e, the unstable antenna CM03 is manually flagged because of noisy amplitudes. In EB uid://A002/Xff99e1/X24be0, antennas CM04 and CM06 are manually flagged from scan 23 to 29 due to unstable amplitudes not allowing for a proper phase referencing calibration. Data for 10 baselines in EB uid://A002/X107376c/X6126 are manually flagged due to high phase noise rms. The QA score for this stage (#14 in the weblog) is still showing the value provided before the flagging. Therefore the QA score of 0.10 is not representative. For more information on the added manual flags, please visit the stage #2 hifa_flagdata in the weblog.

Imaging issues: None. Note: Strong continuum detection occurs.

It is recommended that the PI carefully assess the results on the hif_findcont weblog page, and the "line-free moment 8" images on the cube imaging weblog page.

Self-calibration was not performed.

This dataset has been checked for the so-called "renormalization issue". A detailed description about this issue, and how the renormalization process is carried out, can be found at the following link: <https://help.almascience.org/kb/articles/what-are-the-amplitude-calibration-issues-caused-by-alma-s-normalization-strategy>. The requirement of rescaling due to any astronomical lines detected in this dataset has been evaluated and the largest peak rescaling value has been estimated to be 3.0%, as displayed in the table in the hifa_renorm task of the delivered weblog. The effect of rescaling is a channel-dependent increase of the line flux, which is largest in the brightest channels. This is not an increase in flux-scale uncertainty, but a flux offset correction. When comparing such an offset to the nominal absolute flux accuracy, which is 5% in Bands 3, 4 & 5 and increasing to 20% in Bands 9 & 10, it is concluded that offsets up to 2% are considered negligible. Since the rescaling factor is above this threshold, this dataset has been corrected for the above issue.

QA2 was performed on the image whose channel width is the one closest to the representative bandwidth selected by the PI in ALMA OT, in the spectral window closest in frequency to the representative frequency. This scheduling block was declared QA2 SEMIPASS, meaning that it did not meet the PI requested performance in RMS sensitivity. The SB is delivered anyway after attempting to add 2 more execution blocks to the already achieved execution count. The dynamic range reached is 25 and the achieved RMS

did not decrease after the additional EBs. The PI is encouraged to attempt self-cal:
https://casaguides.nrao.edu/index.php?title=First_Look_at_Self_Calibration

Representative Window -
Image name: uid____A001_X2d20_X2e71.s34_0.HD14055_sci.spw22.mfs.l.iter1.image
Robust = 0.50
Beam size = 4.557 x 4.067 arcsec
RMS = 0.243 mJy/beam over 1836.204 MHz
RMS at PI requested resolution = 0.120 mJy/beam over 7500.000 MHz (6514.61 km/s)

Aggregate Continuum -
Image name: uid____A001_X2d20_X2e71.s36_0.HD14055_sci.spw16_18_20_22.cont.l.iter1.image.tt0
Robust = 0.50
Beam size = 4.641 x 4.141 arcsec
RMS = 0.102 mJy/beam over 6.379 GHz

For additional information on the calibration and imaging pipeline products please see the Knowledgebase article:
<https://help.almascience.org/kb/articles/what-calibration-and-imaging-products-will-be-delivered-to-me>

RMS and beam size at representative frequency					
Sensitivity goal	0.087 [mJy] over bandwidth 7.500 [GHz]				
Angular resolution goal	N/A				
Achieved RMS					
for desired bandwidth	0.120 [mJy]		for continuum		0.100 [mJy]
Synthesized beam	Mean (arcsec)		4.305		
Major axis (arcsec)	4.557	Minor axis (arcsec)		4.067	Position angle (deg)
					-5.684

Execution blocks summary													
EB	N Ant.	Start Time	End Time	ToS (sec)	Avg. Elev. (deg)	Trans. Elev.	Mean PWV (mm)	Phase RMS (deg)*	Min BL (m)	Max BL (m)	AR (")	MRS (")	EF
uid://A002/Xff99e1/X1b0c4	11	2022-10-16 03:40:	2022-10-16 05:25:	6257	32.6	33.1	0.0	9.117	8.9	48.9	3.3	19.2	1.16
uid://A002/Xff99e1/X24be0	11	2022-10-17 06:13:	2022-10-17 07:55:	6147	0.0	0.0	0.0	19.026	8.9	48.9	3.3	19.2	0.78
uid://A002/Xff99e1/X2b2ee	11	2022-10-18 03:45:	2022-10-18 05:29:	6258	32.8	33.1	0.0	18.285	8.9	48.9	3.3	19.2	1.12
uid://A002/Xff99e1/X2c3bb	11	2022-10-18 06:00:	2022-10-18 07:43:	6149	26.0	33.1	0.0	13.354	8.9	48.9	3.3	19.2	1.16
uid://A002/Xfff3d8/Xd46	11	2022-10-19 04:25:	2022-10-19 06:07:	6098	32.4	33.1	0.0	12.173	8.9	48.9	3.3	19.2	1.16
uid://A002/Xffde1/X2b1a	10	2022-10-20 04:16:	2022-10-20 05:57:	6111	32.6	33.1	0.0	11.857	8.9	48.9	3.3	19.2	1.16
uid://A002/Xffde1/X3835	10	2022-10-20 05:58:	2022-10-20 07:42:	6194	25.2	33.1	0.0	11.065	8.9	48.9	3.3	19.2	1.08
uid://A002/Xffde1/X9bd6	10	2022-10-21 05:05:	2022-10-21 06:47:	6119	0.0	0.0	0.3	8.347	8.9	91.0	1.9	13.3	0.39
uid://A002/X1001fdc/X2085	11	2022-10-23 03:31:	2022-10-23 05:12:	6104	32.9	33.1	0.0	10.534	8.9	48.9	3.3	19.2	1.16
uid://A002/X1001fdc/X29ae	11	2022-10-23 05:12:	2022-10-23 06:56:	6232	28.6	33.1	0.0	9.684	8.9	48.9	3.3	19.2	1.16
uid://A002/X1001fdc/Xc06e	10	2022-10-24 03:02:	2022-10-24 04:45:	6213	32.4	33.1	0.0	25.320	8.9	48.9	3.3	19.2	1.16
uid://A002/X1003af4/X3a2c	11	2022-10-25 04:03:	2022-10-25 05:29:	5137	32.8	33.1	0.0	9.895	8.9	48.9	3.3	19.2	1.16
uid://A002/X1003af4/X14e08	11	2022-10-27 03:29:	2022-10-27 05:10:	6110	32.9	33.1	0.0	10.923	8.9	48.9	3.3	19.2	1.16

EB	N Ant.	Start Time	End Time	ToS (sec)	Avg. Elev. (deg)	Trans. Elev.	Mean PWV (mm)	Phase RMS (deg)*	Min BL (m)	Max BL (m)	AR (")	MRS (")	EF
uid://A002/X1006d51/X228e	10	2022-10-29 03:11:	2022-10-29 04:55:	6220	0.0	0.0	0.0	10.789	8.9	48.9	3.3	13.5	0.70
uid://A002/X101088d/Xbf80	11	2022-12-23 01:43:	2022-12-23 03:26:	6233	25.6	33.1	0.0	22.188	8.9	48.9	3.3	19.2	0.77
uid://A002/X105e6e0/X7c5e	10	2023-04-21 16:26:	2023-04-21 18:10:	6237	32.1	33.1	0.0	47.141	8.9	48.0	3.3	18.3	0.52
uid://A002/X107376c/X6126	9	2023-05-16 14:37:	2023-05-16 16:24:	6416	32.4	33.1	0.0	65.434	8.9	48.9	2.6	16.4	0.71
uid://A002/X107fa96/X9a88	10	2023-06-01 14:26:	2023-06-01 16:10:	6242	29.6	33.1	0.0	26.088	8.9	48.0	3.3	18.3	1.18
uid://A002/X107fa96/X150f6	10	2023-06-03 13:27:	2023-06-03 15:11:	6214	32.4	33.1	0.0	51.192	8.9	48.9	3.3	18.3	1.06
uid://A002/X1086384/X10bc1	9	2023-06-10 13:37:	2023-06-10 15:21:	6263	30.6	33.1	0.4	17.720	8.9	91.0	2.1	16.4	1.31
uid://A002/X1088045/X3c7b	10	2023-06-11 11:44:	2023-06-11 13:28:	6235	32.1	33.1	0.0	15.303	8.9	48.0	3.3	18.3	1.50

*WVR-corrected value averaged over all basebands and scans.

Pipeline executions

CASA version	Pipeline version	Report date	Processing time
6.4.1.12	2022.2.0.68	2023-06-20 11:07:48	1 day, 15:10:51

Calibrator source fluxes

Source name	Flux (Jy)	Flux error (Jy)	Frequency (GHz)	Intents
J0237+2848	0.927	0.093	331.308	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.924	0.092	333.200	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.909	0.091	343.250	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.902	0.090	345.141	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.793	0.079	331.308	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.801	0.080	333.199	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.787	0.079	343.249	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.786	0.079	345.141	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.872	0.087	331.307	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.876	0.088	333.199	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR

Source name	Flux (Jy)	Flux error (Jy)	Frequency (GHz)	Intents
J0237+2848	0.864	0.086	343.249	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.865	0.086	345.140	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.829	0.083	331.307	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.834	0.083	333.199	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.829	0.083	343.249	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.824	0.082	345.140	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.776	0.078	331.307	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.779	0.078	333.198	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.771	0.077	343.248	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.771	0.077	345.140	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.748	0.075	331.306	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.745	0.074	333.198	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.732	0.073	343.248	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.730	0.073	345.139	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.850	0.085	331.306	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.851	0.085	333.198	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.832	0.083	343.247	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.825	0.083	345.139	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.836	0.084	331.306	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.835	0.084	333.197	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.826	0.083	343.247	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.821	0.082	345.139	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.764	0.076	331.305	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.759	0.076	333.196	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.746	0.075	343.246	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR

Source name	Flux (Jy)	Flux error (Jy)	Frequency (GHz)	Intents
J0237+2848	0.748	0.075	345.138	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.835	0.084	331.305	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.834	0.083	333.196	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.822	0.082	343.246	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.820	0.082	345.138	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.756	0.076	331.304	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.758	0.076	333.196	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.742	0.074	343.246	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.742	0.074	345.137	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.811	0.081	331.304	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.808	0.081	333.195	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.798	0.080	343.245	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.796	0.080	345.137	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.901	0.090	331.303	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.893	0.089	333.194	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.884	0.088	343.244	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.881	0.088	345.136	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.731	0.073	331.301	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.734	0.073	333.193	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.725	0.073	343.243	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.718	0.072	345.134	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.610	0.061	331.273	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.602	0.060	333.165	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.599	0.060	343.214	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.597	0.060	345.106	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR

Source name	Flux (Jy)	Flux error (Jy)	Frequency (GHz)	Intents
J0237+2848	0.513	0.051	331.290	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.510	0.051	333.182	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.510	0.051	343.231	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.483	0.048	345.123	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.520	0.052	331.303	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.548	0.055	333.195	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.534	0.053	343.245	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.538	0.054	345.136	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.579	0.058	331.311	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.584	0.058	333.203	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.566	0.057	343.253	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.577	0.058	345.144	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.541	0.054	331.312	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.532	0.053	333.204	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.532	0.053	343.254	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.528	0.053	345.145	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.563	0.056	331.315	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.567	0.057	333.207	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.556	0.056	343.257	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.554	0.055	345.148	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.593	0.059	331.316	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.593	0.059	333.207	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.580	0.058	343.258	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR
J0237+2848	0.578	0.058	345.149	CALIBRATE_ATMOSPHERE, CALIBRATE_PHASE, CALIBRATE_POINTING, CALIBRATE_WVR

Spectral Windows

Transition	Central Frequency (sky, GHz)	Bandwidth (GHz)	N of channels
13CO (SW-1)	331.297	2.000	4096
Continuum_2 (SW-1)	333.189	2.000	128
_Continuum_1 (SW-1)	343.239	2.000	128
CO_v=0_3-2 (SW-1)	345.130	2.000	4096

Instructions

INTERFEROMETRY INSTRUCTIONS OVERVIEW

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#####

INTRODUCTION

ALMA currently provides data products for each member observation unit set (MOUS), corresponding to one or more executions of a single scheduling block (SB). In all cases except for polarization projects, each execution is independently calibrated, and then all executions are imaged together.

The following text describes the contents of this data package. Further details and updates can be found in the ALMA Knowledge Base (KB) articles indicated throughout this document.

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DATA DELIVERY CONTENTS

Each data package tarball expands to contain a single member.<ous_id> directory with the following sub-directories: calibration, script, qa, log, product. (Note: The ALMA archive will have prepended many product and script names with the MOUS UID, compared to the names displayed in the pipeline's weblog.)

- 'qa/' contains a number of files that summarize the quality of the observations and products. There is a "QA0 report" for each execution giving an overview of the conditions (Tsys, phase RMS) and quality of each observation. The "QA2 report" (this file) provides an overall assessment of the complete observation and final products. For pipeline-processed data there is a member.<ous_id>*.weblog.tgz file for each time the pipeline was run (calibration, imaging, or calibration+imaging) that unpacks into "weblog" (open the pipeline-<DATE>/html/index.html file in a web browser), which is a series of webpages that present the details of the pipeline run, many diagnostic plots, and quality metrics. For manually calibrated products, several text files and png images are provided instead, which can be used in conjunction with the *scriptForCalibration.py described below to understand how the data were calibrated.

- 'product/' contains fits files of the imaging products. In some cases, only a subset of all science targets/spectral windows are imaged, to demonstrate the quality of the calibration and images. If the products were created by the imaging pipeline, a detailed comprehensive explanation of what images were created, how continuum was subtracted, and the quality of the processing is contained in the pipeline weblog (see above). If images were created manually, please refer to the *scriptForImaging.py described below for details of the process.

- 'script/' contains the scripts to:

- calibrate raw visibilities (*casa_pipescript.py in the case of pipeline calibration, and *scriptForCalibration.py in the case of manual calibration).

- restore calibrated visibilities without fully rerunning the pipeline (*casa_piperestorescript.py)

- restore calibrated visibilities ("*scriptForPI.py"; works for either pipeline or manually calibrated data - see the section "How to restore the calibrated Measurement Set (MS) for your data" further below).

- create images (*casa_pipescript.py in the case of pipeline imaging, and *scriptForImagingPrep.py and *scriptForImaging.py in the case of manual imaging).

- In case the calibration was done by the automated pipeline, you will also see the Pipeline Processing Request file (PPR).

(Note: the ALMA archive may prepend the script names with the MOUS UID.)

- 'calibration/' contains the files needed for calibration, including calibration and flagging tables and manually introduced flagging commands, which allow the user to start from the initial ASDM files and obtain the fully calibrated data.

- 'log' contains the CASA log files.

For more information see the ALMA QA2 Data Products document (<https://almascience.org/processing/qa2-data-products>), the ALMA Technical Handbook (in particular chapter 11 on the QA2 pass criteria), and the relevant Call for Proposals for this Cycle, which are available for download from the ALMA Science Portal at <https://almascience.org/documents-and-tools>.

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INTERFEROMETRIC CALIBRATION AND IMAGING REGENERATION

This section concerns ALMA interferometric (12m Array and ACA) data. The ALMA KB article "How to Restore the Calibrated Measurement Set (MS) for your Data" at <https://help.almascience.org/kb/articles/how-do-i-obtain-a-file-of-calibrated-visibility-measurement-set-for-almadata> will contain an up-to-date version of the procedures described below.

How to Restore the Calibrated Measurement Set (MS) for your Data

Calibration is the first step of processing interferometric data. Many users will want to recreate the calibrated measurement set and then perform the second step (imaging) using their own scripts, or by modifying the imaging script provided with the data.

In order to obtain your calibrated data, download the raw data in ASDM format from the science archive. If you downloaded and untarred all available files for this delivery as described in the notification email, then you will already see these files (with names like "uid*.asdm.sdm") in an additional sub- directory called "raw". Otherwise, you will need to download and untar the tar balls of the raw data in the same directory that you untarred the tarball of the products, creating the "raw" sub- directory and associated ASDM files.

For the next step, you will need the right version of CASA to be installed. Please find the line starting with "CASA version used for reduction:" in your QA2 report or README. The version indicated there is what you need to use for running the scriptForPI.

Once the raw data is in place, cd to the "script" sub-directory, start CASA:

```
casa --pipeline
```

(Note: For some versions, CASA is released without pipeline and the "--pipeline" switch is not available. You will need a version with the pipeline if there is a file named "PPR*.xml" or "*pprequest.xml" in your script/ sub-directory. If there is no such file, you will not be able to run the the pipeline or restore pipeline-calibrated data. For more information on the execution of the pipeline, please refer to the ALMA Science Pipeline User's Guide available at <https://almascience.org/documents-and-tools/pipeline-documentation-archive>).

Then look for the script ending in "scriptForPI.py" and execute it, e.g.:

```
execfile("member.uid___A001_X123a_X45.scriptForPI.py")
```

Running the scriptForPI will result in one or more calibrated measurement sets (MSs) ready for imaging. You can use the uid_*.ms directly, keeping in mind that the calibrated data is stored in the CORRECTED data column, and all sources (including calibrators) are in the MS. If you want to run the imaging pipeline, you should run it from this point, without changing the uid_*.ms.

Imaging

If the data were imaged with the pipeline, many details of what images were created, how and why, are

contained in the pipeline weblog. Each "task" of the pipeline weblog corresponds to a pipeline "hif_" or "hifa_" task which the user can run manually to reproduce what the pipeline did, optionally changing parameters. The full sequence of such tasks run by the pipeline are contained in "script/*casa_pipescript.py", which could be run to completely reproduce all pipeline processing. Alternatively, the calibrated measurement set produced above can be used as the basis for running only the imaging tasks in the pipeline. See https://casaguides.nrao.edu/index.php/ALMA_Imaging_Pipeline_Reprocessing for details.

Each imaging section of the weblog also links to the CASA commands that were used by the pipeline these form a good initial template which users can modify to manually image the data with different parameters.

If the data were manually imaged, refer to "script/*scriptForImaging.py", which contains the CASA commands that were used to create the image products from the calibrated MS. The "scriptForImaging.py" may partially be interactive (for masking) and should be executed by copy and paste.

Options

1) If you want to force scriptForPI.py to split out the science spectral windows, you need to set the variable DOSPLIT=True before starting the scriptForPI in CASA.

2) The scriptForPI will usually run *casa_piperestorescript.py, which applies the calibration and flagging tables from the calibration/ sub-directory to the raw MS, restoring a calibrated MS. If *casa_piperestorescript.py is not available (as is the case for manually-calibrated datasets), the scriptForPI will instead run the entire calibration script, re-creating the calibration and flagging tables. You can force the execution of *casa_pipescript.py to rerun the entire calibration pipeline (instead of only restoring the existing calibration) by moving *casa_piperestorescript.py out of the script directory. Rerunning the calibration can be useful if you want to tweak its parameters. Otherwise the restore is faster.

3) The scriptForPI offers some global variables for your convenience. You can find an explanation in this Knowledgebase Article: <https://help.almascience.org/kb/articles/what-global-variables-are-available-in-scriptforpi-py>

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PRIMARY BEAM CORRECTION

The images included in delivery are corrected for the primary beam (PB) response, i.e. the dependence of the instrument's sensitivity on direction within the field of view.

For each image, two files are being delivered:

- a) the PB-corrected image (file name ending in ".pbcor.fits")
- b) the image of the PB which was used in the correction (ending in ".pb.fits" or ".flux.fits", typically gzipped in which case you need to unzip the file before using it)

The image noise was measured in the uncorrected image. The corrected image (a) was then obtained by dividing the uncorrected image by the PB image (b). The uncorrected image can be recovered using the CASA task impbcor in mode "m":

```
impbcor(image='image.pbcor.fits', pbimage='image.pb.fits', mode='m', outfile='image.recovered')
```

See also the ALMA KB article "Where is the Primary Beam Correction Information in my Delivered Data" at

<https://help.almascience.org/kb/articles/where-is-the-primary-beam-correction-information-in-my-delivered-data> for any updates to this procedure.

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